



COVENANT UNIVERSITY
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P.M.B 1023, OTA, OGUN STATE, NIGERIA.

TITLE OF EXAMINATION: ALPHA SEMESTER EXAMINATION
COLLEGE: SCIENCE AND TECHNOLOGY
DEPARTMENT: COMPUTER AND INFORMATION SCIENCES
SESSION: 2018/2019
COURSE CODE: CSC 213
COURSE TITLE: STRUCTURED PROGRAMMING
INSTRUCTION: ANSWER ANY FOUR QUESTIONS

SEMESTER: ALPHA
CREDIT UNIT: 3

Time: 2 1/2 Hours

1. a) With a given integer number n (user supplied), write a python program to generate a dictionary that contains $\{i : i*i\}$ such that i is an integer number between 1 and n (both included). Then the program should print the dictionary. (3.5 marks)

- b) Given the following dictionary:

```
inventory = {  
    'gold' : 500,  
    'pouch' : ['flint', 'twine', 'gemstone'],  
    'backpack' : ['xylophone', 'dagger', 'bedroll', 'bread loaf']  
}
```

Write a python statement do the following:

- Add a key to inventory called 'pocket' and set the value of 'pocket' to be a list consisting of the string values 'seashell', 'strange berry', and 'lint'.
- Sort the items in the list stored under the 'backpack' key.
- Remove 'dagger' from the list of items stored under the 'backpack' key.
- Add 50 to the number stored under the 'gold' key.
- Print the resulting dictionary

(4 marks)

- c) Study the following dictionaries for an online grocery store:

```
stock = {"banana": 6, "apple": 0, "orange": 32, "pear": 15, "avocado": 1, "strawberry": 4}  
prices = {"banana": 4, "apple": 2, "orange": 1.5, "pear": 3, "avocado": 5, "strawberry": 8}
```

Using the dictionaries,

- (i) Write a function **compute_bill** that takes one argument (which is a list of shopping items) as input. The function should return the total price of the all items in the shopping list. While you loop through each item of the list, only add the price of the item to the total price if the item's stock count is greater than zero. If the item is in stock, after adding the price to the total, subtract one from the item's stock count.
- (ii) Your program should create a shopping list with any number of the items in the stock dictionary and then calculate the bill for the items in the shopping list. (5 marks)

- d) The following are names of participants in a swimming and bowling event:

Swimming – Eric, Chioma, Bayo, Ekaete, Bridget, Hassan, Amina, Jumoke

Bowling – Hassan, Ngozi, Chris, Jide, Tosin, Eric, Zoe, Halima, Bayo, Ebuka

Write a python program to create a set for each event. Your program should;

- Print out list of participants who attended both events.
- Print out list of participants who attended only one of the events.
- Print out list of participants who attended only the swimming event but not the bowling event.
- Print out list of participants who attended only the bowling event but not the swimming event.
- Print out list of all participants.

(5 marks)

2. a) Mention and briefly describe the two types of files (6marks)
 b) Write a function called **middle** that takes a list as parameter and returns a new list that contains all but the first and last elements. (3.5marks)
 c) Suppose a file named **ChapelServiceAttendance.txt** contains the headers FName, LName, student number and four chapel attendance records for each student in the month of November. "1" signifies presence while "0" means the student was absent. A tab ('\t') delimits the data:

FName	LName	Student number	3Nov	10Nov	17Nov	24Nov
Nnamdi	Chukwu	12CF1122	1	1	0	1
Aderopo	Ude	12CF2233	0	0	1	1
Shina	Kudi	11CA4431	1	1	1	1
Helen	Mike	12CD8877	0	1	1	1
Kyle	Tunde	12CV7766	0	0	0	1

Write a python program to read the content of the file and display the attendance percentage for all students on the screen. (Hint: compute Attendance percentage as $\text{No. of times present} / 4 * 100$)

e.g. Nnamdi Chukwu 12CF1122 75%
 Aderopo Ude 12CF2233 50%

(8marks)

3. a) Briefly describe the following terms and give suitable examples where applicable (7.5marks)
 (i) Lists (ii) Tuples (iii) Dictionary (iv) Sets (v) File

b) Given

tuple = ('abcd', 787, 2.24, 'john', 70.2)
 tinytuple = ('abcd', 786, 2.23, 'john', 70.2)

What is the output of:

- print(tuple[0]) (1mark)
- print(tuple[1:3]) (1mark)
- print(tuple[2:]) (1mark)
- print(tinytuple[-4]) (1mark)
- print(tinytuple[-3:]) (1mark)

c) Write a Boolean function named **is_prime** which takes an integer as an argument and returns True if the argument is a prime number, or False otherwise. Use the function in a program that prompts the user to enter a number and then displays a message indicating whether the number is prime. (5marks)

4. a) Mention and describe any three string testing methods you know (6marks)
 b) A car's kilometer-per-gallon (KPG) can be calculated with the following formula:

KPG - Kilometer driven / Gallons of gas used

Write a program that asks the user for the number of kilometers driven and the gallons of gas used. It should calculate the car's KPG and display the result. (4marks)

c) Write a Python program to print the even numbers from a given list called **list1** to a new list called **list2**. (4.5marks)

Sample List : [1, 2, 3, 4, 5, 6, 7, 8, 9]

Expected Result : [2, 4, 6, 8]

d) Using **for** loop and **range()** in Python, write a program to print the following sequence:

3 6 9 12 15 18 21 24

(3marks)

5. a) List three escape characters and their effects in Python (4.5marks)
 b) Mention and briefly describe the three steps to be taken when a file is used by a program (9marks)
 c) Write a python program to calculate the sum of the first 20 terms of the series.
 $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots$ (4marks)

6. a) Given the list of some books of the Bible

```
BibleBooks = ['Matthew', 'Mark', 'Luke', 'John', 'Psalms', 'Romans']
```

Use List methods to achieve the following:

- Remove **Psalms** from the list. (1mark)
- Get the index of **Luke**. (1mark)
- Insert **Acts** in the list after **John** (1mark)
- Expand the list by adding **Corinthians** to the List (1mark)
- Sort the list (1mark)
- What will be the output of print (**BibleBooks**)? (1mark)

- b) Write a Python program to accept a string as input and add 'ing' at the end of the given string (length should be at least 3). If the given string already ends with 'ing' then add 'ly' instead. If the length of the given string is less than 3, leave it unchanged. For example:

(6.5marks)

Sample String (Input)	Expected Result (Output)
'ab'	'ab'
'abc'	'abcing'
'string'	'stringly'

- c) Write a Python program that writes to file called **products.txt**. The following records should be written to the product file. (5marks)

PName	Price	Category	Manufacturer
Gizmo	N19.99	Gadgets	GizmoWorks
Powergizmo	N29.99	Gadgets	GizmoWorks
SingleTouch	N149.99	Photography	Canon
MultiTouch	N203.99	Household	Hitachi